



# LIGHTING PRACTICE: ENVIRONMENTAL CONSIDERATIONS FOR OUTDOOR LIGHTING

AN AMERICAN NATIONAL STANDARD





**ANSI/IES LP-11-20**

**LIGHTING PRACTICE:  
ENVIRONMENTAL CONSIDERATIONS  
FOR OUTDOOR LIGHTING  
AN AMERICAN NATIONAL STANDARD**

Publication of this Recommended Practice  
has been approved by IES.  
Suggestions for revisions  
should be directed to IES.

**Prepared by  
The IES Outdoor Environmental Lighting Committee**



*Copyright 2020 by the Illuminating Engineering Society.*

*Approved by the IES Standards Committee February 13, 2020 as a Transaction of the Illuminating Engineering Society.*

*Approved April 21, 2020 as an American National Standard.*

*All rights reserved.* No part of this publication may be reproduced in any form, in any electronic retrieval system or otherwise, without prior written permission of the IES.

Published by the Illuminating Engineering Society, 120 Wall Street, New York, New York 10005.

IES Standards are developed through committee consensus and produced by the IES Office in New York. Careful attention is given to style and accuracy. If any errors are noted in this document, they should be forwarded to Brian Liebel, Director Standards, at [standards@ies.org](mailto:standards@ies.org) or the above address for verification and correction. The IES welcomes and urges feedback and comments.

Printed in the United States of America.

ISBN# 978-0-87995-354-6

---

## **DISCLAIMER**

IES publications are developed through the consensus standards development process approved by the American National Standards Institute. This process brings together volunteers representing varied viewpoints and interests to achieve consensus on lighting recommendations. While the IES administers the process and establishes policies and procedures to promote fairness in the development of consensus, it makes no guaranty or warranty as to the accuracy or completeness of any information published herein.

The IES disclaims liability for any injury to persons or property or other damages of any nature whatsoever, whether special, indirect, consequential or compensatory, directly or indirectly resulting from the publication, use of, or reliance on this document.

In issuing and making this document available, the IES is not undertaking to render professional or other services for or on behalf of any person or entity. Nor is the IES undertaking to perform any duty owed by any person or entity to someone else. Anyone using this document should rely on his or her own independent judgment or, as appropriate, seek the advice of a competent professional in determining the exercise of reasonable care in any given circumstances.

The IES has no power, nor does it undertake, to police or enforce compliance with the contents of this document. Nor does the IES list, certify, test or inspect products, designs, or installations for compliance with this document. Any certification or statement of compliance with the requirements of this document shall not be attributable to the IES and is solely the responsibility of the certifier or maker of the statement.

## **AMERICAN NATIONAL STANDARD**

Approval of an American National Standard requires verification by ANSI that the requirements for due process, consensus, and other criteria have been met by the standards developer.

Consensus is established when, in the judgment of the ANSI Board of Standards Review, substantial agreement has been reached by directly and materially affected interests. Substantial agreement means much more than a simple majority, but not necessarily unanimity. Consensus requires that all views and objections be considered, and that a concerted effort be made toward their resolution.

The use of American National Standards is completely voluntary; their existence does not in any respect preclude anyone, whether that person has approved the standards or not, from manufacturing, marketing, purchasing, or using products, processes, or procedures not conforming to the standards.

The American National Standards Institute does not develop standards and will in no circumstances give an interpretation to any American National Standard. Moreover, no person shall have the right or authority to issue an interpretation of an American National Standard in the name of the American National Standards Institute. Requests for interpretations should be addressed to the secretariat or sponsor whose name appears on the title page of this standard.

**CAUTION NOTICE:** This American National Standard may be revised at any time. The procedures of the American National Standards Institute require that action be taken to reaffirm, revise, or withdraw this standard no later than five years from the date of approval. Purchasers of American National Standards may receive current information on all standards by calling or writing the American National Standards Institute.

**Prepared by: The IES Outdoor Environmental Lighting Committee**

**Bob Parks, *Chair***  
**David M. Keith, *Secretary***

**Members**

|                 |               |                |                           |
|-----------------|---------------|----------------|---------------------------|
| N. E. Clanton   | T. Longcore   | B. S. Renouf   | Y. Tyukhova               |
| P. K. Ericson   | T. K. McGowan | C. Shattuck    | K. M. Zielinska-Dabkowska |
| R. Gibbons      | G. A. Ortt    | J. Slade       |                           |
| M. Hartley      | D. W. Paulin  | J. M. Stockman |                           |
| R. L. Henderson | R. Rainer     | C. S. Stubbe   |                           |

**Advisory Members**

|                |               |                  |             |
|----------------|---------------|------------------|-------------|
| C. L. Bailey   | J. Frazer     | A.-M. E. Lemeiux | M. Pattison |
| M. Desajardins | F. J. Krahe   | D. Lenasi        | M. M. Yeung |
| R. Farrar      | G. F. Kruggel | J. Leung         |             |

---

# CONTENTS

|            |                                                            |           |
|------------|------------------------------------------------------------|-----------|
| <b>1.0</b> | <b>Introduction and Scope</b>                              | <b>1</b>  |
| 1.1        | Introduction                                               | 1         |
| 1.2        | Scope                                                      | 1         |
| <br>       |                                                            |           |
| <b>2.0</b> | <b>Environmental and Health Considerations</b>             | <b>1</b>  |
| 2.1        | Effects of Exterior Lighting on Human Health               | 2         |
| 2.2        | Effects of Exterior Lighting on Animals                    | 3         |
| 2.3        | Effects of Exterior Lighting on Plants                     | 4         |
| <br>       |                                                            |           |
| <b>3.0</b> | <b>Minimizing the Effects of Light Generated by Humans</b> | <b>5</b>  |
| 3.1        | Light Pollution                                            | 5         |
| 3.1.1      | Natural Phenomena                                          | 6         |
| 3.1.2      | Effect of Electric Lighting                                | 6         |
| 3.1.3      | Sky Glow Models                                            | 6         |
| 3.1.4      | Special Considerations                                     | 6         |
| 3.1.5      | Controlling Light Pollution                                | 7         |
| 3.2        | Light Trespass                                             | 8         |
| <br>       |                                                            |           |
| <b>4.0</b> | <b>Lighting Zones</b>                                      | <b>9</b>  |
| 4.1        | Lighting Zone Definitions                                  | 10        |
| 4.2        | Factors to be Considered in Developing Lighting Zones      | 12        |
| 4.2.1      | Additional Design Considerations for Lighting Zones        | 13        |
| <br>       |                                                            |           |
| <b>5.0</b> | <b>Developing a Lighting Ordinance</b>                     | <b>13</b> |
| 5.1        | Getting Started                                            | 14        |
| 5.2        | Model Lighting Ordinance Details                           | 14        |
| 5.2.1      | Purpose, Intent, and Preamble                              | 14        |
| 5.2.2      | General Regulations for All Exterior Lighting              | 14        |
| 5.2.3      | Establish Lighting Zones                                   | 15        |
| 5.2.4      | Establish Curfews                                          | 15        |
| 5.2.5      | Specific Regulations for Nonresidential Exterior Lighting  | 16        |
| 5.2.6      | Specific Regulations for Residential Lighting              | 16        |
| 5.2.7      | Lighting by Special Permit Only                            | 16        |
| 5.3        | Approval Requirements                                      | 17        |

**6.0    The Effects of Luminaire and Sign Luminance on Perceptions of the Environment ..... 17**

**References ..... 19**





## 1.0 Introduction and Scope

### 1.1 Introduction

The introduction of exterior lighting will often have a profound effect on the natural world. It may change complex ecosystems in ways that are not immediately obvious or easily discernable. For this reason, the first question should always be, is lighting necessary? This should be followed by, what is the specific task to be lighted, how much luminance is required, and for what duration? When designing exterior lighting the environmental effects should always be considered. Lighting is also a sustainability issue, since the energy consumed often comes from power generation that produces its own environmental impacts. While exterior lighting may be necessary for mobility, the feeling of safety, and commerce, used indiscriminately, the light pollution produced by it reduces the enjoyment of the natural nighttime sky, and may create biodiversity issues (see *ANSI/IES LP-10-20, Lighting Practice: Sustainable Lighting – An Introduction to the Environmental Impacts of Lighting*<sup>1</sup>).

These concerns include:

- The addition of electric lighting into the nocturnal environment may generate serious negative ecological consequences to a wide range of species.
- Exterior lighting at night may contribute to human health issues like sleep disruption,<sup>2</sup> obesity,<sup>3</sup> diabetes,<sup>4,5</sup> cancer,<sup>6,7</sup> and depression.<sup>8</sup>
- Blue-rich white light, because of greater atmospheric scattering, disproportionately increases sky glow, but all exterior lighting can increase night sky brightness.
- Light pollution can have severe negative effects on astronomical research.

### 1.2 Scope

This Lighting Practice (LP) document outlines the environmental considerations of exterior lighting, especially as related to glare, sky glow, light trespass and the impact of electric light at night on flora and fauna. In addition, this LP provides information on how to assign lighting zones, and how to use the *Joint IDA-IES Model Lighting Ordinance (MLO) with User's Guide*, as a

basis for an exterior lighting ordinance.<sup>9</sup> Finally, this LP discusses community-based design and makes specific recommendations for lighting outdoor areas.

A number of IES Recommended Practice documents (RPs) provide design guidelines for specific exterior lighting applications. This LP is not intended to supersede those documents. Where conflicting uses and desires for the nighttime environment arise, the parties involved should work together to resolve the issues by reviewing all of the relevant criteria to reach a consensus about which criteria will guide the final resolution.

## 2.0 Environmental and Health Considerations

Humans have been using nighttime lighting since the dawn of history. However, since the industrial revolution the amount and distribution of this additional light has increased dramatically. Satellite images of the night sky from the U.S. National Oceanic and Atmospheric Administration's National Geophysical Data Center show vividly how heavily illuminated the planet is (see **Figure 2-1**).



**Figure 2-1. The United States at night from space.** (Image courtesy of NASA/NOAA)

Light pollution is of special concern to astronomical observatories, and the negative effects of exterior lighting on astronomical observations are well documented and can effect research at great distances

from the source of the light pollution. Most of the major observatories in the United States—for example, Palomar and the Kitt Peak Observatory (KPNO)—are affected to a degree that observations of faint objects are significantly compromised. Sky glow near KPNO is produced by the lights of Tucson (56 miles away), along with Phoenix, about 120 miles to the North. The Mauna Kea observatories on the big island of Hawaii now see an increasing number of lights from other Hawaiian islands. Professional and amateur astronomers' observatories are least harmed by lighting using fully-shielded luminaires (with no uplight), long-wavelength light sources, and minimal illuminance (see **Figure 2-2**). Any plans for electric light use near observatories should take these constraints into account.



**Figure 2-2. Fully shielded streetlight, with pedestrian scale poles.** (Image courtesy of Jim Richardson)

Research data clearly show the deleterious effects, including profound negative environmental consequences, of electric lighting in the nocturnal environment.<sup>10-18</sup> Ecological light pollution includes chronic or periodically increased illumination, unexpected changes in illumination, and direct glare. Light pollution has demonstrable effects on the behavioral and population ecology of organisms in natural settings. As a whole, these effects derive from changes in orientation, disorientation, or misorientation, as well as attraction or repulsion from the altered light environment, which in turn may affect foraging, reproduction, migration, and communication.

If a project is in an environmentally sensitive area, such as a wilderness preserve, the question may include a straightforward requirement to have the

lighting make no adverse impact on the surroundings. The first question should always be, why does this need to be lighted? Users entering such areas have few expectations for modern conveniences and may be more willing to take responsibility for their own actions. Difficult access may also restrict the number of people in such an area and may limit the user group to those who share a similar respect for nature and for other people and their property.

While more research is required to fully understand the interaction between electric light sources and animals and plants, the overall negative impact is not in doubt. The examples cited here are intended as representative of ongoing research. When development is considered in environmentally sensitive areas, the use and application of nighttime lighting should be carefully reviewed.

At the most basic level, the way to do this is to remove unnecessary luminaires, turn off luminaires except when needed, reduce the amount of light, block light from entering sensitive areas, and direct light only where it is needed. For example, in general, the equivalent of full moonlight approximately 0.1 lux (0.01 fc) average<sup>19</sup> should be considered adequate unless there are specific requirements for higher levels of illumination in these sensitive areas. Using lighting controls provides the flexibility to reduce or extinguish the lighting during times when people are less likely to be present. Local regulations may specify curfews that require lighting to be reduced after close of business or other set times.

## 2.1 Effects of Exterior Lighting on Human Health

Research concerning non-visual human responses to optical radiation has raised questions about the effects of exterior lighting on human health. As with any research concerning human health, the information is complex and difficult to place in context. This is due in part to the complexity of investigations, the lack of experimental control, the ability of results to be generalized, and finite resources that make it impractical to explore all relevant hypotheses. IES TM-18-18 represents the current knowledge of the IES Light and Human Health Committee on the known relationships between optical radiation and human health.<sup>20</sup>

IES TM-18-18 indicates that exposure to optical radiation\* affects human physiology and behavior, both directly (acute effects including melatonin suppression, elevated cortisol production, increased core temperature) and indirectly (resetting the internal circadian body clock). Some of the uncertainty about the research in this area arises from the differences in spectral sensitivity between the visual and nonvisual responses to optical radiation.

Since there are differences in spectral sensitivity between the visual and nonvisual responses to optical radiation, existing units of measurement based on the photopic luminous efficiency function (for definition, refer to *ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering*<sup>21</sup>) are not adequate to characterize nonvisual responses, such as melatonin suppression. The position of the IES is that new spectral weighting functions that are appropriate for characterizing nonvisual response(s) to optical radiation are needed.

Older and more recent measurement systems have not incorporated all key aspects of the optical radiation stimuli: quantity, spectrum, duration, timing, direction, and exposure history. Until such time as a new metric is available to quantify responses to these optical radiation stimuli, there is uncertainty about what effects should be considered harmful to health, and for which populations; melatonin suppression and circadian disruption are clearly variable, depending on the individual and the characteristics of exposure(s). In addition, most of the research has occurred in laboratories; it is of interest to the IES to complement such studies with those that characterize the effect of optical radiation on circadian, neuroendocrine, and neurobehavioral responses in field applications. At present, there is inadequate documentation of what constitutes typical exposures to optical radiation in exterior settings after sunset that would constitute real-world normative data.

Optical radiation detected by the retina impacts an individual's behavior, psychology, and perception of the environment. The position of the IES is to promote and

encourage a more complete understanding of human responses to optical radiation leading to improved designs for all lighted environments through additional research, with specific emphasis on additional field research to document typical exposures to optical radiation in exterior settings.

## 2.2 Effects of Exterior Lighting on Animals

Exterior lighting has impacts on animals, from insects to large predators. It disrupts normal breeding, predation and predator avoidance, migration, feeding, communication, and social interactions—in short, every aspect of animal life.<sup>22,23</sup> Almost all small rodents and carnivores, 80% of marsupials, and 20% of primates are nocturnal.<sup>21</sup> These species can be affected by lighting, as can diurnal species that are disrupted during resting periods or extend activity into nighttime. Lighting in sensitive areas should be limited spatially, of minimal duration, directed only where needed, and reduced in intensity as much as possible (no more than full moonlight as a guideline). In addition, it is important to consider how the spectral power distribution affects the sensitive species.<sup>24,25</sup>

For example, along the U.S. Atlantic coast from North Carolina south, sea turtle hatchlings are genetically programmed to use starlight and moonlight reflections off the water to guide them from their nests to the water (see **Figure 2-3**). Any electric light source brighter than natural light can confuse the hatchlings and lead them inland, away from the ocean.<sup>26</sup> Local codes in this region may require the use of shielded luminaires, amber or red light, controls, and other limitations on exterior nighttime lighting, as well as shielding of interior lighting, to mitigate the presence of confusing light sources.



**Figure 2-3 Baby sea turtles.** (Image courtesy of Mark Sully/Shutterstock)

\* Optical radiation: Electromagnetic radiation having wavelengths between approximately 100 nm and 1 mm. Source: *ANSI/IES LS-1-20, Nomenclature and Definitions for Illuminating Engineering*.<sup>21</sup> Online: [www.ies.org/standards/definitions/](http://www.ies.org/standards/definitions/). (Accessed 2019 Jun 26).

Migratory birds are also confused by nighttime lighting. Nocturnally migrating birds are attracted to lighted areas at the landscape scale,<sup>27</sup> to individual lighted structures,<sup>27</sup> and to lighted windows.<sup>28</sup> As an example, along the shoreline of the Great Lakes, lighting has caused birds to fly into the sides of buildings. Shielding interior lights, using flashing instead of constant lights, and reducing or limiting exterior nighttime lighting, especially during periods of peak migration, may help reduce species loss.<sup>29</sup>

In addition, electric lighting has been shown to disrupt numerous natural behavioral patterns, including<sup>18</sup>:

- Foraging behavior and increased predation
- Biological clocks, which could affect light-stimulated behaviors, such as mating success and group-mediated anti-predator vigilance
- Increased mortality from roadkill
- Dispersal movements and wildlife corridor use
- Breeding behavior in birds, including nest site selection, increased predation, and timing of breeding
- Possible support for invasive species (mostly insects)
- Alteration of territorial behavior
- Delays and changes in migratory behavior
- Reduced populations of nocturnal insects, which

has a cascade effect on their predators, and on pollination (see **Figure 2-4**)

### 2.3 Effects of Exterior Lighting on Plants

Plants have evolved a wide range of photoreceptors that perceive and respond to light signals, particularly in the ultraviolet, blue, red, and near-infrared regions of the electromagnetic spectrum.<sup>30</sup> These many photoreceptors allow the plant to measure and respond to four parameters of the light environment: spectral quality, intensity, direction, and duration. Electric light at night disrupts these natural patterns and has direct and indirect effects on wild plants.<sup>31</sup>

Electric light at night can be bright enough to affect the timing of plant growth, growth form, and resource allocation (see **Figure 2-5**). Some of the many processes sensitive to light are:

- Seed germination<sup>18</sup>
- Stem elongation<sup>32</sup>
- Leaf expansion<sup>18</sup>
- Flower development<sup>33</sup>
- Fruit development<sup>18</sup>
- Bud dormancy (when leaves stop growing)<sup>34</sup>
- Leaf senescence<sup>35</sup>
- Leaf drop<sup>35</sup>



**Figure 2-4 White-lined sphinx moth on Seedskadee National Wildlife Refuge**  
(Image courtesy of Tom Koerner/USFWS)

Plants that are sensitive to photo-periodicity in their flowering, bud dormancy, or leaf senescence in response to day length may be adversely affected by prolonged light. They need sunlight during the day and darkness at night to sustain a healthful life. Landscape lighting that is left on continuously at night is seldom necessary and should be avoided.

Indirect effects can occur through adverse impacts on pollinators; this then affects pollination rate and seed production.<sup>36</sup> Mitigating these effects involves reduction of light and control of portions of the spectrum most disruptive for pollinators.<sup>37</sup>

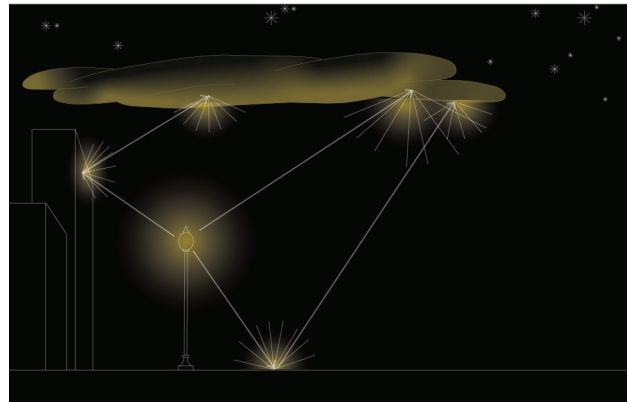


**Figure 2-5.** This photograph shows an example of delayed leaf drop which is the most prevalent impact of street lights on trees; it causes increased mortality and stunted growth. The trees shown are both chestnut trees. The one on the left exhibits normal leaf drop, while the one on the right, which is much closer to the streetlight, is experiencing a delay. (Image courtesy of Lamiot<sup>38</sup>)

### 3.0 Minimizing the Effects of Light Generated by Humans

*Light pollution*, *sky glow*, and *obtrusive light* are terms used to describe the excess or nuisance light created by humans. *Light pollution* and *light trespass* (see **Figure 3-1**) have become extremely important considerations whenever a new exterior lighting design is being prepared. As people increasingly appreciate the beauty and benefits of the night, they have become less tolerant of unnecessary and intrusive lighting. (Refer to *ANSI/IES LP-10-20, Lighting Practice: Sustainable Lighting – An Introduction to the Environmental Impacts of Lighting*<sup>1</sup>

for a further discussion of light pollution and light trespass issues and design considerations.)



**Figure 3-1.** Light pollution and light trespass impacts from a non-shielded luminaire. (Image redrawn from original by Clanton & Associates)

#### 3.1 Light Pollution

*Light pollution* is the combination of all the adverse or obtrusive effects of electric light that produces artificial sky glow:

- Unnecessary, unwanted, or wasted light
- Light that damages or degrades the nighttime environment
- Light that negatively impacts humans and other species

*Sky glow* is the brightening of the night sky that results from the scattering and reflection of light from the constituents of the atmosphere (gaseous molecules and aerosols), in the direction of the observer. It has two components: natural sky glow and artificial sky glow (see **Figure 3-2**).<sup>21</sup>



**Figure 3-2** Clouds over Chicago reflecting light from the city below. (Image courtesy of Jim Richardson)

Sky glow deprives urban residents of the opportunity to see the night sky as their ancestors did, stargaze, and relax under a beautiful natural night sky. It can hamper professional and amateur astronomers' attempts to view the night sky through telescopes. Sky glow is increased by electric lighting at night.

**3.1.1 Natural Phenomena.** Natural sources of sky glow exist, mostly due to the energy reradiated by atmospheric chemicals during nighttime cooling (known as airglow). The spectral compositions of these natural phenomena are well documented and are typically addressed by professional astronomers through the use of specialized filters and processing techniques.

Two light-scattering processes that contribute to sky glow are scattering of light by, for example, nitrogen and oxygen molecules (called *Rayleigh scattering*), and scattering by aerosols (called *Mie scattering*). Rayleigh scattering is stronger for short-wavelength (blue) light than longer-wavelength light, while scattering from aerosols is little affected by wavelength. Atmospheric aerosols are more concentrated over urban environments, increasing the impact of electric lighting on perceived and measurable sky glow compared to more rural environments.

**3.1.2 Effect of Electric Lighting.** Electric lighting increases sky glow above natural background levels through a combination of direct and reflected light. Any light emitted above the horizontal contributes directly to sky glow. Light reflected from the ground or from vertical surfaces can contribute to sky glow, depending on the ground cover conditions (snow is highly reflective, for example).

Mesopic and scotopic vision are dominant at lower levels of illumination. At these times, the eye perceives short-wavelength (blue) light more easily than long-wavelength (red). Historically, exterior lighting for roadways and large areas has primarily utilized high pressure sodium, which has minimal short-wavelength light. Low pressure sodium is used occasionally, which has no short wavelength light. Blue-rich light sources (such as high-CCT LED and metal halide) provide broader spectral content, but the increase in short-wavelength light increases the relative impact of Rayleigh scattering, and thus can lead to relatively higher sky glow.

**3.1.3 Sky Glow Models.** In an urban center at night it may be that only a few stars are visible to the naked eye. At a site tens of kilometers distant from any electric light sources, thousands of stars can be seen overhead, as well as the Milky Way. The horizon, however, is brightened over nearly all of Europe and the USA. Sky glow on the horizon has been documented even at locations hundreds of kilometers away from large cities.<sup>39</sup>

Sky glow levels may be measured with suitable photometers<sup>40</sup> or estimated using radiative transfer models. While numerous models have been used in the past, there are currently three main models. The model of Falchi and Cinzano was used to produce a "World Atlas" of night sky brightness at zenith on clear nights.<sup>38</sup> A model by Kocifaj and Kundracik<sup>41</sup> was used by the U.S. Department of Energy to analyze the impact of changes to LEDs and can model the whole sky dome, including under overcast skies.<sup>42</sup> The model of Aubé is the most physically detailed, and takes into account local terrain.<sup>43</sup>

Sky glow models are based on satellite data and calibrated using radiometers, both of which are challenged by the shift toward the use of white LEDs. Finally, relatively little work has been done on understanding sky glow on cloudy nights. Therefore, although the conclusions from the existing models are robust, especially for clear nights, additional experimental and theoretical work is warranted for a future in which LEDs become increasingly widespread.

**3.1.4 Special Considerations.** In areas near astronomical observatories, and in areas with endangered species or special conditions such as turtle nesting areas, protection from sky glow is a primary concern (see **Figure 3-3**). Fully shielded luminaires (with no uplight) should be specified, with designs that use the minimum possible lighting levels. Such features as path lights emitting red light work well to preserve dark adaptation. Reflectorized markings rather than electric lighting should be used on roadways whenever possible. Nearly monochromatic light sources (such as low pressure sodium or amber LEDs) are advantageous for exterior areas because any unwanted light that enters observatory instruments often can be filtered out. The use of sources with limited spectral composition does



**Figure 3-3. Sky glow from Los Angeles significantly degrades scientific research at Griffith Observatory.** (Image courtesy of Wikimedia.org<sup>44</sup>)

not completely eliminate atmospheric light pollution from the telescope's view, but it helps greatly and is highly recommended for such locales.

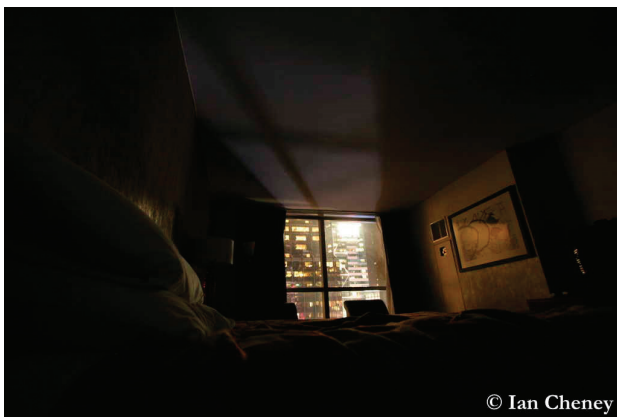
**3.1.5 Controlling Light Pollution.** The methods that best control light pollution are:

- *Minimize upward emissions.* Exterior lighting systems, including lighting for sports activities, parking lots, buildings, landscape, storage areas, vehicle sales lots, and other exterior retail areas or drive-up windows should be designed to minimize or eliminate direct upward emission. It is also important to note that near-horizontal emissions, either direct or reflected, have a much greater effect on sky glow than does light output at angles more than 10 degrees below horizontal. Emissions into natural habitats should be avoided.
- *Minimize non-target lighting.* Lighting systems that project light upward, such as architectural and sign lighting, should be designed to minimize emission of light that does not illuminate the target area (see *ANSI/IES RP-39-19, Recommended Practice: Off-Roadway Sign Luminance*<sup>45</sup>).
- *Do not over-light.* Use no higher a lighting level than is needed for the expected visual tasks.
- *Turn off or dim exterior lighting during low or no use.* All exterior lighting, including advertising sign lighting, should be turned off after use unless necessary for safety and security. Usually, safety and security illuminance can be at levels much lower than those needed when the area is in use.
- *Turn off interior high-rise office building lighting* when not in use, or use window treatments to prevent it from exiting the building.
- *Use no higher correlated color temperature (CCT) than necessary* for color appearance of intended tasks. Low CCT is recommended to reduce astronomical light pollution<sup>42</sup> and to reduce negative impacts on wildlife.<sup>37</sup> A maximum CCT of 3000 K for area lighting in commercial zones is recommended, while CCTs lower than 2200 K should be used to minimize impacts in sensitive environments. *Note:* While CCT is often used to describe white light sources, it does not accurately characterize the blue content that can increase the perception of glare and may negatively impact ecology. For reference, light sources with CCT of 3000 K or lower will typically have less than 25% spectral power distribution (SPD) below 530 nm. *Important:* This is just what is necessary to avoid increasing negative impacts over existing HPS spectral characteristics.<sup>37,42</sup>

The reader is referred to the *Joint IDA-IES Model Lighting Ordinance (MLO)* for limitations on sky glow.<sup>9</sup>

### 3.2 Light Trespass

The topic of light trespass relates to light that is obtrusive off site. A typical example is the “light shining in my window” complaint (see **Figure 3-4**). Solutions include shielding and/or reducing the luminance of the offending luminaire so that its brightness is not directly visible from the adjacent property, turning off the light after curfew, or in some cases, eliminating the light source if not required. The offending illumination is a form of obtrusive light.\*



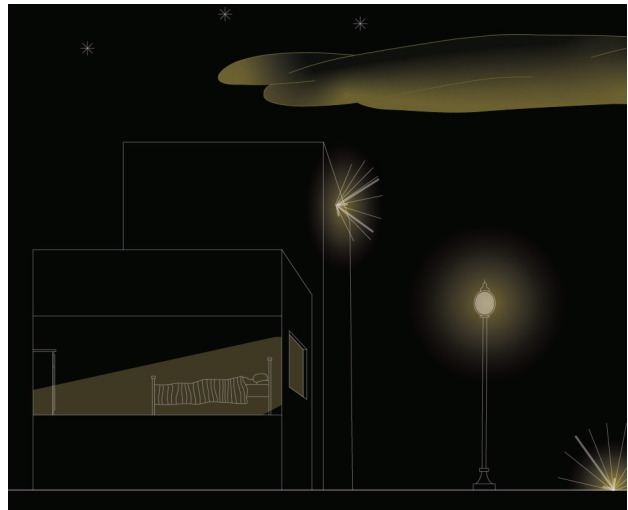
**Figure 3-4. An example of light trespass.** (Image courtesy of Ian Cheney)

Light trespass usually happens when an adjacent property receives unwanted light (see **Figure 3-5**). This does not necessarily mean high illuminance levels. For example, a sports light a block or so away might prove annoying shining in a window, even though the illuminance produced is actually rather low.

General suggestions to help control light trespass problems include:

- Inspect areas adjacent to the lighting design location to identify and consider any potential problems involving residences and other places of sleep, as well as roadways, airports, and topographical challenges, such as a luminaire located higher on a hillside than a bedroom window.

\* Obtrusive light: Direct or reflected light that, because of quantitative, directional, or spectral attributes in a given context, causes annoyance, discomfort, distraction, or a reduction in the ability to see. Source: ANSI/IES LS-1-20, *Nomenclature and Definitions for Illuminating Engineering*.<sup>21</sup> Online: [www.ies.org/standards/definitions/](http://www.ies.org/standards/definitions/). (Accessed 2019 Jun 26).



**Figure 3-5. An illustration of light trespass. Decorative globe allows light to spill onto the façade and into residential windows.** (Image redrawn from original by Clanton & Associates)

- Select luminaires with tightly controlled intensity (candela) distributions that meet the BUG rating (Backlight, Uplight, and Glare—refer to ANSI/IES TM-15-20, *Technical Memorandum: Luminaire Classification System for Outdoor Luminaires*<sup>46</sup>) for specific lighting zones.
- For non-shielded luminaires, use low-lumen light sources to minimize brightness.
- Contain light within the design area by carefully selecting, locating, and mounting the luminaires.
- Keep aiming angles for façade lighting low so that the entire beam always falls within the intended area during (and after) the design and installation process. Aim luminaires away from sensitive areas.
- Conform to the Model Lighting Ordinance, described in **Section 5**, or applicable local zoning ordinances.<sup>9</sup>

Efforts have been made in numerous jurisdictions to write ordinances or bylaws controlling light trespass. These ordinances have met with a range of success, depending on practical enforcement. One method to control light trespass is based on using specific lighting zone descriptions (see **Section 4.0**) that then underlie any restrictions. (Refer to the *Joint IDA-IES Model Lighting Ordinance (MLO)*<sup>9</sup> and ANSI/IES TM-15-20<sup>46</sup> for limits on luminaire BUG ratings, property line

maximum illuminance, and lumen density by lighting zone.) Lighting zones and the BUG rating system can provide the basis for restricting the brightness that will be tolerated in a specific environment. Within any category, a curfew time may be established, allowing higher lighting levels if necessary during those hours when the curfew is not in effect.

It is important to note that since light trespass is extremely subjective, there is no single set of values or limits that will work in every situation. Therefore, while these recommendations serve to reduce serious light trespass, their implementation is not a guarantee against objections. In some situations, such as a sports field in a small park closely surrounded by residences, no methods and combinations of lighting design, aiming, or control can provide for both safe play and satisfy some neighbors' desires for limited light trespass. Consensus solutions involving field locations, curfews to restrict the hours of nighttime use, glare abatement, and/or landscape screens should be reached by all the parties involved.

### 4.0 Lighting Zones

---

Zoning is a well-established practice in community planning. The fundamental idea behind zoning is that it allows a community to determine and regulate appropriate types of use in different areas within its jurisdiction—for example, to define acceptable land uses in different areas. Lighting zones, which reflect the base (or ambient) light levels desired by a community, work well with land use zones in setting limits on the type and amount of lighting that may be used in different areas. Using lighting zones allows a great deal of flexibility and customization without the burden of excessive regulation.

The choice of an appropriate lighting zone is a matter of judgment based on community priorities for any given area. It is recommended that the lowest reasonable lighting zone(s) be adopted. Selection of a lighting zone or zones should be based not on existing conditions but rather on the type of lighting environments

the jurisdiction seeks to achieve. For instance, new development on previously rural or undeveloped land may be zoned as LZ-1.

Lighting zones are best implemented as an overlay to the established zoning, especially in communities where a variety of zone districts exist within a defined area or along an arterial street. Where zone districts are cohesive, it may be possible to assign lighting zones to established land use zoning. It is recommended that the lighting zone includes churches, schools, parks, and other uses embedded within residential communities.

Lighting zones help communities minimize the contrast (and conflict) between extremes in lighting, such as a brightly illuminated car dealership adjacent to or within sight of a residential neighborhood, or an illuminated sports facility in the middle of a residential neighborhood. Lighting zones may also determine restrictions on exterior lighting that impact “places of sleep,” such as residential areas, hospitals, hotels, and long-term care facilities. Lighting zones may also employ vertical distinctions, such as in mixed-use facilities where the commercial aspects are on the street with residential units on the higher levels. Lighting zones also encourage minimal changes in visual adaptation when traveling from site to site.

However, if an adjacent use could be adversely impacted by allowable lighting, the authority having jurisdiction may require that a particular site meet the requirements for a lower lighting zone. For example, the authority could specify Lighting Zone 1 or 2 requirements if a commercial development were adjacent to a residence, hospital, or open space, or to any land assigned to a lower zone.

Community involvement is important in determining lighting zone issues, such as whether and how to light churches, schools, ice rinks, or playing fields. Among the factors that should be considered are neighborhood ambient conditions, lighting expectations, special environmental concerns, and how interior lighting may affect the outdoor environment. Curfews and controls are appropriate for all lighting zones and should be specified.

#### 4.1 Lighting Zone Definitions

Because identifying the appropriate outdoor lighting zone is a matter of judgment and consensus, there is no absolute means of determining which lighting zone designation is appropriate for a given area. The same type of lighting application may fall into different lighting zones in different jurisdictions or using different standards. As used in the *Joint IDA-IES Model Lighting Ordinance (MLO)*,<sup>9</sup> the lighting zones are defined with suggested uses:

- **LZ-0: No ambient light**

LZ-0 (see **Figure 4-1**) includes areas where the natural environment could be seriously and adversely affected by small amounts of electric lighting at night. This includes biological cycles of flora and fauna, and human enjoyment and appreciation of the natural environment. The vision of human residents and users of these areas is adapted to the total darkness, and they do not expect to see electric lighting. Human activity is sparse and is subordinate in importance to the natural environment. There is no expectation for electric lighting. Although some lighting is allowed, it is required to be controlled.

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| LZ-0 | Lighting Zone 0 should be applied to areas in which permanent lighting is not expected and when used, is limited in the amount of lighting and the period of operation. LZ-0 typically includes undeveloped areas of open space, wilderness parks and preserves, areas near astronomical observatories, or any other area where the protection of a dark environment is critical. Special review should be required for any permanent lighting in this zone. Some rural communities may choose to adopt LZ-0 for residential areas. | Recommended default zone for wilderness areas, parks and preserves, and undeveloped rural areas. Includes protected wildlife areas and corridors. |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|

**Figure 4-1. LZ-0 description from the MLO.**

- **LZ-1: Low ambient light**

LZ-1 (see **Figure 4-2**) includes developed areas within a natural environment and areas of human activity that are inherently dark at night. Electric lighting at night could adversely affect the biological cycles of flora and fauna, or could interrupt the quiet, dark character of the area. The vision of human residents and users of these areas is adapted to the low light levels, and they do not expect to see electric lighting except where absolutely necessary to improve visibility and safety. In these limited areas, low light levels

are appropriate. Lighting is expected to be non-continuous (i.e., pools of light rather than uniform lighting along a path or roadway). After curfew, both light levels and uniformity may be reduced in some areas. An example of a parking lot in an LZ-1 area is shown in **Figure 4-3**.

|      |                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                  |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LZ-1 | Lighting Zone 1 pertains to areas that desire low ambient lighting levels. These typically include single and two family residential communities, rural town centers, business parks, and other commercial or industrial/storage areas typically with limited nighttime activity. May also include the developed areas in parks and other natural settings. | Recommended default zone for rural and low density residential areas. Includes residential single or two family; agricultural zone districts; rural residential zone districts; business parks; open space include preserves in developed areas. |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Figure 4-2. LZ-1 description from the MLO.**



**Figure 4-3. A parking lot located in a Lighting Zone 1 community.** (Image courtesy of Bob Parks)

- **LZ-2: Moderate ambient light**

LZ-2 (see **Figure 4-4**) includes areas human activity (i.e., habitation, recreation, and/or work) where electric lighting may be required for increased mobility and convenience at night. The vision of human residents and users of these areas is adapted to moderate light levels, and they have moderate expectations of electric lighting. Lighting is expected to be non-continuous (e.g., pools of light at crosswalks or intersections, rather than uniform lighting along a path or street). After curfew, both light levels and uniformity may be reduced in some areas as activity levels decline. **Figure 4-5** shows an example of a street located in an LZ-2 area.

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                   |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>LZ-2</b></p> | <p>Lighting Zone 2 pertains to areas with moderate ambient lighting levels. These typically include multifamily residential uses, institutional residential uses, schools, churches, hospitals, hotels/motels, commercial and/or businesses areas with evening activities embedded in predominately residential areas, neighborhood serving recreational and playing fields and/or mixed use development with a predominance of residential uses. Can be used to accommodate a district of outdoor sales or industry in an area otherwise zoned LZ-1.</p> | <p>Recommended default zone for light commercial business districts and high density or mixed use residential districts. Includes neighborhood business districts; churches, schools and neighborhood recreation facilities; and light industrial zoning with modest nighttime uses or lighting requirements.</p> |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Figure 4-4. LZ-2 description from the MLO.



Figure 4-5. A street located in an LZ-2 area: fully shielded lighting, uniform distribution. (Image courtesy of Bob Parks)

- **LZ-3: Moderately high ambient light**

LZ-3 (see **Figure 4-6**) includes areas of human activity (i.e., habitation, recreation, and/or work) where electric lighting may be continuous and is required for and convenience at night. The vision of human residents and users of these areas is adapted to moderately high light levels, and they have moderate to high expectations of electric lighting. Lighting is expected to be continuous (e.g. lighting delivered fairly evenly along the length of a path or street). After curfew, both light levels and uniformity may be reduced in some areas as activity levels decline. **Figure 4-7** shows an example of building façade lighting in an LZ-3 area.

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                        |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>LZ-3</b></p> | <p>Lighting Zone 3 pertains to areas with moderately high lighting levels. These typically include commercial corridors, high intensity suburban commercial areas, town centers, mixed use areas, industrial uses and shipping and rail yards with high night time activity, high use recreational and playing fields, regional shopping malls, car dealerships, gas stations, and other nighttime active exterior retail areas.</p> | <p>Recommended default zone for large cities' business district. Includes business zone districts; commercial mixed use; and heavy industrial and/or manufacturing zone districts.</p> |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Figure 4-6. LZ-3 description from the MLO.



Figure 4-7. A long-term care facility in an LZ-3 area: well-shielded lighting—no direct uplight. (Image courtesy of David Roederer)

- **LZ-4: High ambient light**

LZ-4 (see **Figure 4-8**) includes areas of high levels of human activity at night, including significant interaction among pedestrians and/or vehicles. The vision of humans when outside is typically adapted to moderate light levels. Lighting is continuous and is required for safety and convenience. Expectations for electric lighting are high, both in terms of light levels and uniformity along pathways or streets. However, both light levels and uniformity may be reduced after curfew hours in some areas as activity levels decline. **Figure 4-9** shows an example of an urban entertainment area designated as LZ-4.

|                    |                                                                                                                                                                                                                                                                                                 |                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <p><b>LZ-4</b></p> | <p>Lighting zone 4 pertains to areas of very high ambient lighting levels. LZ-4 should only be used for special cases and is not appropriate for most cities. LZ-4 may be used for extremely unusual installations such as high density entertainment districts, and heavy industrial uses.</p> | <p>Not a default zone. Includes high intensity business or industrial zone districts.</p> |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|

Figure 4-8. LZ-4 description from the MLO.



Figure 4-9. Example of an entertainment district in an LZ-4 urban area. (Image courtesy of Bob Parks)

## 4.2 Factors to be Considered in Developing Lighting Zones

The following questions may be addressed whenever the use of lighting zones is proposed.

- *What is the purpose behind the adoption of a lighting zone to regulate exterior lighting?* Is it to preserve or to improve the present state of exterior lighting in the area? If it is the former, then selecting the appropriate lighting zone is a matter of matching the zone to the prevailing lighting. If it is the latter, some care is necessary in selecting the lighting zone to be sure that the hoped-for result can be achieved. This means paying attention to the arrangements for exemptions, waivers, relaxations, and “grandfathering.”
  - *Is the intention to adopt one lighting zone for the whole jurisdiction, or to match the lighting zones to the land-use zones?* The approach will depend upon the diversity of land use in the jurisdiction. Adopting the same outdoor lighting zone for the whole jurisdiction is appropriate where there is little diversity of land use or nighttime activity. Where there is a wide diversity of land use and nighttime activity, matching the lighting zones to the land-use zones is a more sensible approach.
  - *What is the desired activity level of the area covered by the proposed lighting zone?* Selection of a lighting zone or zones should be tailored to the type of lighting environments the jurisdiction seeks to achieve. In general, the lowest level of lighting appropriate for the activity level is best, rather than allowing for the occasional or extreme use. For example, a school playing field adjacent to residential areas could be zoned the same as the residential area, with provision for occasional exceptions for night games. Façade lighting on sporting facilities, churches, and other buildings that may have occasional nighttime use can be allowed to be energized at all times, but required to be dimmed after curfew.
  - *What are the relative priorities for light pollution, public safety, developing the local economy, and preserving local wildlife?* Community priorities will suggest different lighting zones. For example, LZ-0 can be assigned to areas where there is no expectation for nighttime lighting. It should also be applied to areas sensitive to wildlife, such as shore areas for sea turtles during breeding and hatching time.
- Lower lighting zones should be applied to places where sleeping will occur. In a rural or suburban area where there are fewer late nighttime activities and a decrease in late nighttime traffic, priority can easily be given to reducing the light pollution impact on a nature preserve by adopting zones LZ-1 or LZ-2, and by adopting curfews. Communities that value the nighttime sky as an asset may choose LZ-0 or LZ-1. Conversely, in an urban area where traffic density is high, where there is more late nighttime activity, or where crime is a problem, higher levels of lighting may be very appropriate for applying zone LZ-3. Zone LZ-4 should be reserved for areas of continuous nighttime activity, such as an urban entertainment district.
- *What are the relative priorities to be given to reducing sky glow, limiting glare, diminishing light trespass, and conserving energy?* The answer to this question is related to the size of the lighting zone. Giving priority to reducing sky glow implies that the lighting zone or zones should be low and cover as large an area as possible, because all lighting installations over a large area contribute to sky glow. Giving priority to reducing glare and light trespass means that the lighting zone or zones should be tailored to the zoning code because glare and light trespass are local phenomena, specific to the topography of the situation. Lighting should also be reduced during periods of low activity or with seasonal conditions such as snow-covered terrain. This is referred to as *adaptive lighting* and allows for lighting level reductions. In addition, any reduction in exterior lighting will conserve energy.
  - *Are there any special circumstances regarding the area covered by the proposed outdoor lighting zone?* Special circumstances come in three forms: circumstances that imply a high priority to one of the objectives listed above, areas that are exempt from local regulations, and areas that are illuminated to a much higher level than most of the surrounding area but only for a limited time. An example of the first is a professional observatory. The presence of such an observatory nearby demands that priority be given to minimizing light pollution. An example of both the first and the second is a prison. State and federal prisons are exempt from local regulations, and their function requires that the lighting priority be

providing good visibility. An example of the third is an illuminated façade at an outdoor sports stadium. Such sports stadiums are usually illuminated at very high illuminance levels to facilitate play and television transmission, but perhaps only for a short time, so these can be dealt with by means of a curfew.

- *Will a curfew be adopted?* A curfew changes the lighting criteria for the lighting zone at a particular time of night, representing a change in priorities. A curfew should always be considered because most of the benefits of exterior lighting are only felt when there are people present. For example, the lighting of a supermarket parking lot is often changed at midnight. Before midnight, there are many people and vehicles about, so public safety has priority. After midnight, there are significantly fewer people about and vehicles are parked closer to the buildings, so the risk to public safety is reduced and priority can be given to reducing light pollution and trespass in surrounding neighborhoods. Parking lots not normally used at night, such as business parks or commercial buildings, may allow significantly reduced lighting levels. Motion sensors and intelligent control systems can increase and decrease lighting levels as activity varies. A preferred option is to dim the lighting instead of turning off every other light, in order to preserve lighting uniformity.
- *Will exemptions from the requirements be allowed in the lighting zone, and will exceptions be available for special events or installations?* It is up to the jurisdictions to establish exceptions for events and installations that can be brought within the outdoor lighting zone and are appropriate for the area.
- *Will the criteria associated with a lighting zone be relaxed where special visual needs are evident, such as the entrance at a senior housing facility?* The entrance and drop-off area may require higher lighting levels, yet glare is still undesirable for seniors. Sleeping areas still need very low light levels in order to minimize circadian cycle issues.
- *Are adjacent outdoor lighting zones compatible?* Here, compatible means that there is not more than one change in outdoor lighting zone class between adjacent zones; i.e., the zones adjacent to an LZ-2 area should be either LZ-1 or LZ-3, but not LZ-4. This

restriction is necessary because light from one zone will overlap into the adjacent zone. If the difference between adjacent zones is too great, the zone with the stricter criteria should regulate.

**4.2.1 Additional Design Considerations for Lighting Zones.** After the lighting zones are established, additional issues may need to be addressed in the design of the lighting within these zones. These items can be added to the ordinance or addressed within the designs of these areas.

- *Should restrictions on spectral distribution be required in a lighting ordinance?* Currently, the MLO does not address spectral distribution, but a community may want to add restrictions such as requiring the use of monochromatic or amber lighting, especially if located near an observatory or for wildlife concerns such as turtle nesting areas. Some communities may want to provide guidance on spectral distribution and color rendering in order to provide a cohesive community theme and provide better security lighting.
- *Should sign lighting be regulated?* Communities can regulate sign lighting and changing copy duration. Intensity, movement, and luminaire locations for externally illuminated signs are methods in which regulations occur. For internally lighted signs, intensity during daylight hours may also be regulated. Additional guidance may be found in *ANSI/IES RP-39-19, Recommended Practice: Off-Roadway Sign Luminance*.<sup>45</sup>
- *What will be done about lighting that contributes to light pollution, visual clutter, or reduced visual acuity, but is not normally considered a general lighting installation?* This includes illuminated signs and buildings with façade lighting. These installations can still be regulated and controlled within the lighting zones.

## 5.0 Developing a Lighting Ordinance

An effective method to control the quality of exterior lighting at the community level is to use an effective lighting ordinance. Depending upon the locale, authority for the creation and/or implementation of such an

ordinance may or may not rest at the local level. Unless mandated otherwise by a higher political jurisdiction, municipalities, e.g., cities, boroughs, townships, and towns, are well advised to take responsibility for developing and maintaining such an ordinance. Without it, the municipality will have little or no authority to guard against lighting that is excessive, insufficient, abusive, unsafe, or inconsistent with the accepted lighting practices and the best interests of the community.

Such an ordinance performs several important functions. It sets a community standard. By its existence, it educates the community and all its components about the value of quality nighttime lighting. It adds value to the community and its citizens. By having minimum acceptable standards, the municipality can evaluate proposed lighting installations against the standards and require compliance, thus protecting against unpleasant surprises. The ordinance will also accord the community the authority to require remedial action to correct nonconforming lighting installations. In addition, such regulations will be very useful for establishing and maintaining community or neighborhood aesthetic themes.

### 5.1 Getting Started

Preparing an effective lighting ordinance requires forethought and planning. To address the relevant issues, it is helpful to convene a group of interested citizens, business owners, elected or appointed officials, the municipal engineer, the code enforcement officer, representatives from the local police, and other stakeholders. Because lighting may not necessarily be on every elected official's priority list, having representation from the governing body can be helpful in spreading the word. Field trips to experience existing examples of good and bad lighting and to become familiar with local situations and needs are appropriate. Using the *Joint IES-IDA MLO*<sup>9</sup> as a template will help municipalities develop accepted lighting ordinance language and methods of compliance. The municipality's legal representative will be well advised to be involved in the process.

Getting an ordinance approved can be a daunting task and is often a compromise between those who insist upon ironclad control and those who resist any form of regulation. The reality is that if the ordinance language

does not meet the needs of the community, it will not likely be enacted. On the other hand, if the ordinance is enacted but does not meet the needs of those who are responsible for enforcing it, e.g., the municipal engineer and the code enforcement officer, it will likely not get applied. For those reasons, it is prudent to have broad involvement and buy-in by the right stakeholders.

### 5.2 Model Lighting Ordinance Details

Once the issues have been fully explored, the *Joint IDA-IES MLO* may be used as the technical basis for an ordinance.<sup>9</sup> The following sections describe basic considerations that should be incorporated into a new lighting ordinance.

**5.2.1 Purpose, Intent, and Preamble.** The document should first establish the purpose for the ordinance. This can be modified to suit specific community issues. The MLO has a suggested preamble statement that reads:

*The purpose of this Ordinance is to provide regulations for exterior lighting that will:*

- *Permit reasonable uses of exterior lighting that does not exceed the minimum levels specified in IES recommended practices for night-time safety, utility, security, productivity, enjoyment, and commerce.*
- *Minimize adverse offsite impacts of offsite lighting such as light trespass, and obtrusive light.*
- *Curtail light pollution, reduce sky glow and improve the nighttime environment for astronomy.*
- *Help protect the natural environment from the adverse effects of night lighting from gas or electric sources.*
- *Conserve energy and resources to the greatest extent possible.*

### 5.2.2 General Regulations for All Exterior Lighting.

The ordinance needs to establish what exterior lighting is regulated, what is exempt, and what exceptions are allowed. Communities have a choice to regulate lighting on private commercial and/or residential property, as well as lighting in the public right of way, such as street lighting. Governmental properties can be regulated through state or federal regulations. The MLO addresses lighting regulations for private properties, both commercial and residential. The MLO has an optional street lighting ordinance which only

addresses uplighting. Issues such as amortization, grandfathering, and size of property to be regulated are left up to the community. The MLO lists suggested areas of applicability, exemptions, and exceptions that communities can adopt or modify.

Regardless of regulations, the lighting installation is still required to meet local electrical, energy, and applicable sections of building codes.

**5.2.3 Establish Lighting Zones.** Establishing lighting zones is a means to establish different limitations for lighting within a community (see **Section 4**). Some smaller communities may assign only one or two zones to the entire community in order to simplify the regulations. Default zones will typically be LZ-1, LZ-2, and LZ-3. **Section 4.1** provides lighting zone descriptions and suggested lighting zone land use areas. **Figure 5-1** shows an example of lighting zone mapping.

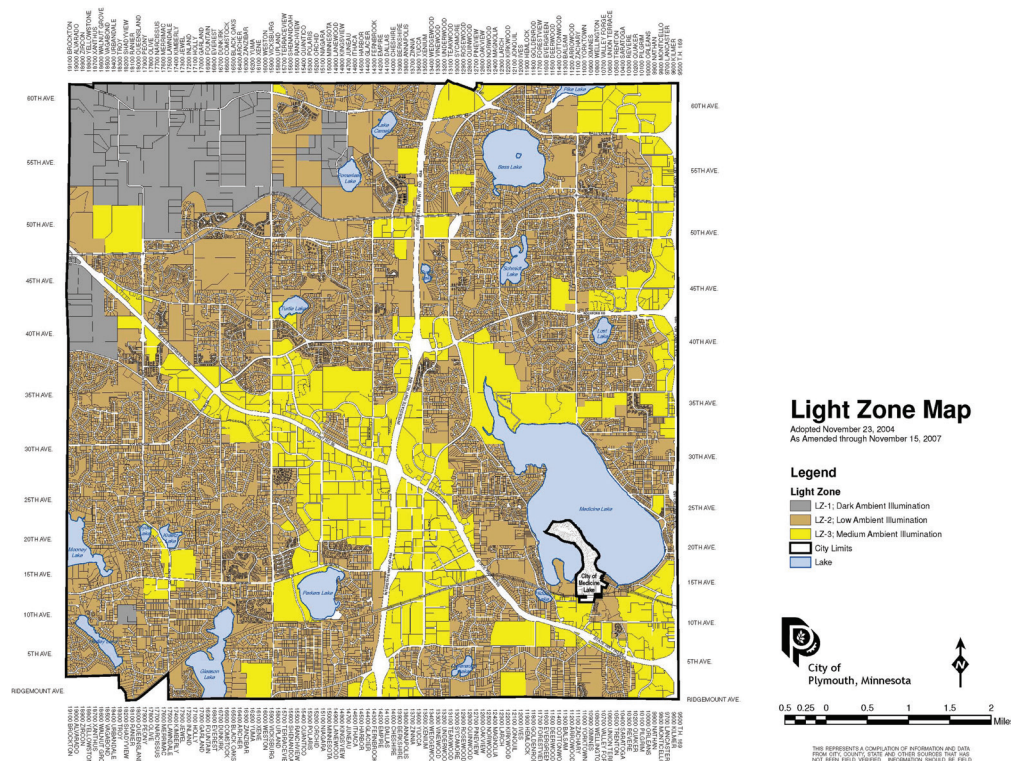
**5.2.4 Establish Curfews.** Curfews are a means of reducing lighting impacts on the community later in the evening or night. Curfews are very important in residential areas and other areas of sleep, as well as in

areas where environmental impacts are potentially high. Street and pedestrian lighting can also be reduced when vehicle and pedestrian activity is reduced. Reduction in street lighting is referred to as *adaptive lighting control*.

During a curfew, exterior lighting will be reduced in intensity by dimming or will be extinguished, through the use of lighting controls. The community may require automatic switching or light-level reduction to comply with curfews.

Exemptions to curfew controls may include stairs, ramps, a single luminaire on the site, or motion sensor controlled luminaires. The safety and security of people and property may be a higher priority on some sites (refer to *IES G-1-16, Security Lighting for People, Property, and Critical Infrastructure* for guidance for these kinds of sites<sup>47</sup>).

During daylight hours, exterior lighting should be extinguished unless it is used for daylight adaptation, such as in parking garage entries, tunnels, and underpasses.



**Figure 5-1. Example of lighting zone mapping in Plymouth, Minn.**

**5.2.5 Specific Regulations for Nonresidential Exterior Lighting.** The Model Lighting Ordinance regulates all exterior lighting for all nonresidential properties.

There are two different methods of compliance: prescriptive and performance. This Lighting Practice (LP) focuses on the prescriptive method, which limits the total site lighting initial luminaire lumens for an individual lighting zone. It is determined by multiplying the area of the improved hardscape by the appropriate lumen density value. It also limits the amount of light from each planned luminaire type. This is done with a Backlight-Uplight-Glare (BUG) rating system (refer to *ANSI/IES TM-15-20, Technical Memorandum: Luminaire Classification System for Outdoor Luminaires*<sup>46</sup>). Zonal lumen limits for luminaires are established per lighting zone. Within a lighting zone, the permitted Backlight rating will change depending on a luminaire's proximity to the property line. **Figure 5-2** shows an example of a service station with luminaires that control the amount of uplight.



**Figure 5-2. Example of service station using fully shielded (no uplight) lighting complying with the MLO.** (Image courtesy of N. Clanton)

Exceptions to this method may include specific applications where lighting is shared among property owners, or low-voltage landscape lighting that is shielded from off site.

### 5.2.6 Specific Regulations for Residential Lighting.

The MLO provides regulations for all exterior lighting in residential properties, including multiple residential properties not having common areas. Since residential property compliance involves the general public, the

regulations are based on the initial lumens of each luminaire. In addition, partial shielding or full shielding is required (see **Figure 5-3**, for example), and each luminaire's light output shall not exceed the specific lumen limits of its lighting zone.



**Figure 5-3. An example of residential lighting in an LZ-0 area.** (Image courtesy of N. Clanton)

*Lamp and light source criteria:* Some jurisdictions permit only health, safety, and welfare issues to be addressed in ordinances. Where permitted, the municipality may wish to establish standards relating to the color appearance of the light, including correlated color temperature and color rendering ability, light source life, and the efficacy of permitted lighting systems. Power quality may also be an issue, such as a desire to limit total harmonic distortion and power factor. Color considerations would be particularly meaningful in retail commercial venues, such as automobile sales facilities. Monochromatic light sources may be appropriate after curfew near observatories.

*Lighting control:* One of the most effective ways to control glare, light trespass, and sky glow and to avoid detrimental effects of light on the environment is to require that offending sources be turned off when not needed or dimmed after curfew. The ordinance should contain a requirement that all non-essential lighting be extinguished or reduced in intensity after curfew or shortly after the close of business.

**5.2.7 Lighting by Special Permit Only.** The ordinance should specify which types of lighting require a special permit. Examples are high-intensity and special-purpose luminaires (such as searchlights and aerial lasers), and

complex and non-conforming uses (such as sporting events, construction, parking facilities, and amusement parks). The ordinance should also specify the permit requirements and process.

### 5.3 Approval Requirements

The lighting ordinance should be specific about submission requirements for review and approval by the municipality. Depending on the compliance method used, they could include:

- Lighting zone and adjacent lighting zone designations
- A lighting layout showing all exterior luminaires, including site- and building-mounted equipment
- Mounting and aiming details
- A luminaire schedule, including initial luminaire lumens and luminaire descriptions
- BUG ratings for each luminaire type
- Site lumen allowance for all improved areas
- Control strategies (all time periods and conditions)
- Light source correlated color temperature and color rendering (if required by the community)
- Plot of maintained illuminance values at grade
- Names of photometric data files used for calculating illuminance values
- Description of on/off control device(s) proposed to control lighting
- Catalog cuts of specified luminaires

Ordinance language should be simple and easy to understand, with relevant terms defined. Clarity and enforceability are key elements. Education of designers, installers, suppliers, code enforcement personnel, and consumers will help ensure the ordinance's success.

## 6.0 The Effects of Luminaire and Sign Luminance on Perceptions of the Environment

The luminances of luminaires and of signs are an important consideration in determining the overall quality of a lighting installation. All luminaires provide

some level of luminance. It is when this luminance becomes excessive that problems will arise. Excessive luminaire or sign luminance can be distracting, uncomfortable, or even visually disabling (see **Figure 6-1**).



**Figure 6-1. Excessive sign brightness from internally illuminated sign.**

The terms *brightness* and *luminance* are often used interchangeably, leading to confusion. ANSI/IES LS-1-20, *Lighting Science: Nomenclature and Definitions for Illuminating Engineering*<sup>21</sup> defines brightness as “the attribute of a visual sensation according to which an area appears to emit more light or less light.” It also has this note under *luminance*: “In common usage, the term *brightness* usually refers to the strength of sensation that results from viewing surfaces or spaces from which light comes to the eye. This sensation is determined in part by the definitely measurable luminance ... and in part by conditions of observation, such as the state of adaptation of the eye.” Due to the subjective nature of a lighting scene, one person may feel a particular luminaire is bright while another person may not. The perception of brightness is affected by the environment in which the luminaire resides, and by the light source’s spectral distribution. Often, the higher the correlated color temperature (CCT), the “brighter” a light source appears. As the background luminance of the scene increases, the apparent brightness of the luminaire becomes lower. Viewing car headlights during the daytime is a good example of a situation involving high luminance but low brightness perception. At night on a dark road, those same headlights may be perceived as overly bright.

Human vision factors also affect the perception of brightness. Blue light is often perceived as brighter than red or green, and can cause more discomfort and disability glare than other, longer wavelengths.<sup>48,49</sup> While red or green light is focused precisely onto the retina, blue light is focused slightly in front of it, which causes a distracting halo around bright blue lights. In addition, blue scatters more widely than other colors as it passes through the eyeball. Also, human vision becomes far more sensitive to blue when ambient light levels are low, a phenomenon known as the Purkinje shift. So a blue light that is merely eye-catching in a brightly illuminated area can become dazzling when the lights are dim.

Because of its subjective nature, brightness is difficult (if not impossible) to measure. However, luminaire luminance and luminous intensity are two photometric quantities correlated with brightness that can be used to specify luminaires. By considering one (or both) of these quantities, a designer can specify a high-quality lighting installation without excessive brightness. The BUG rating system can be used to help select appropriate luminaires (refer to *ANSI/IES TM-15-20, Technical Memorandum: Luminaire Classification System*<sup>46</sup>). In addition, careful selection of an appropriate CCT will help minimize apparent brightness.

Any type of luminaire, whether it is fully shielded or non-shielded, can yield excessive brightness. Factors such as the luminaire optics (size of the luminous area or opening, direct visibility of the light source (clear or diffuse), and the concentration and viewing angle of a luminaire's main light output), the mounting height of the luminaire, the proximity and reflectance characteristics of adjacent objects or surfaces, and the overall luminance of the surrounding scene will all have an effect on how bright a luminaire will appear.

The design of a luminaire will have an effect on its luminance. The higher the luminance, the greater the potential for excessive brightness perception. Poorly designed luminaires producing high-angle light can yield significant glare. As designers move from specifying shielded to non-shielded luminaires (assuming equivalent lumen packages), increasing attention should be paid to luminaire luminance and

the potential for excessive brightness perception. For diffuse sources, average luminance can be calculated from a photometric data file in *ANSI/IES LM-63-19*<sup>50</sup> or *ANSI/IES TM-33-18*<sup>51</sup> format. For directional sources, sample luminaires should be evaluated for appropriate luminance, color, and visual comfort. Care should be given to limit upward light transmission in order to minimize light pollution. Ideally, a mock-up of the scene or direct examination of an already installed site will yield valuable information about the brightness of a specific luminaire.

Additional guidance may be found in *ANSI/IES RP-39-19, Recommended Practice: Off-Roadway Sign Luminance*.<sup>45</sup>

## REFERENCES

- 1 Illuminating Engineering Society. ANSI/IES LP-10-20, Lighting Practice: Sustainable Lighting – An Introduction to the Environmental Impacts of Lighting. New York: IES; 2020.
- 2 Chepesiuk R. Missing the dark: Health effects of light pollution. *Environ Health Perspectives*. 2009 Jan. DOI: <https://doi.org/10.1289/ehp.117-a20>.
- 3 Rybnikova NA, Haim A, Portnov BS. Does artificial light-at-night exposure contribute to the worldwide obesity pandemic? *Inter J Obesity*. 2016;40:815-23. DOI: 10.1515/jbcpp-2016-0116.
- 4 Rakshit K, Thomas AP, Matveyenko AV. Does disruption of circadian rhythms contribute to beta-cell failure in Type 2 diabetes? *Current Diabetes Rep*. 2014;14:474.
- 5 Shi S, Ansari TS, McGuinness OP, Wasserman DG, Johnson CH. Circadian disruption leads to insulin resistance and Obesity. *Current Bio*. 2013;23(5):372-81.
- 6 Zubidat AE, Haim A. Artificial light-at-night – A novel lifestyle risk factor for metabolic disorder and cancer morbidity. *J Basic Clin Physiol Pharmacology*. 2017;28(4).
- 7 Stevens RG, Brainard GC, Blask DE, Lockley SW, Motta ME. Breast cancer and circadian disruption from electric lighting in the modern world. *CA: Cancer J Clinicians*. 2013 Dec 24. DOI: 10.3322/caac.21218.
- 8 Salgado-Delgado R, Osorio AT, Saderi N, Escobar C. Disruption of circadian Rhythms: A crucial factor in the etiology of depression. *Depression Res Treatment*. 2011. DOI: 10.1155/2011/839743.
- 9 International Dark-sky Association (IDA) and Illuminating Engineering Society. Joint IDA-IES Model Lighting Ordinance (MLO) with User's Guide. New York: IES; 2011.
- 10 Allen JA. Destruction of birds by light-houses. *Bulletin Nuttall Ornithological Club*. 1880;5:131-8.
- 11 Gastman EA. Birds killed by electric light towers at Decatur, Ill. *American Naturalist*. 1886;20:981.
- 12 Gaston KJ, Bennie J, Davies TW, Hopkins J. 2013. The ecological impacts of nighttime light pollution: A mechanistic appraisal. *Bio Reviews*. 2013;88:912-27.
- 13 Gaston KJ, Duffy JP, Gaston S, Bennie J, Davies TW. 2014. Human alteration of natural light cycles: Causes and ecological consequences. *Oecologia*. 2014;176:917-31.
- 14 Hölker F, Wolter C, Perkin EK, Tockner K. Light pollution as a biodiversity threat. *Trends Ecol Evol*. 2010;25:681-82.
- 15 Longcore T, Rich C. Ecological light pollution. *Frontiers Ecol Environ*. 2004;2:191-8.
- 16 Longcore T, Rich C. Ecological and Organismic Effects of Light Pollution. Chichester, UK: John Wiley & Sons, Ltd; 2016:1-8.
- 17 McFarlane RW. Disorientation of loggerhead hatchlings by artificial road lighting. *Copeia*. 1963(1):153.
- 18 Rich C, Longcore T, editors. Ecological Consequences of Artificial Night Lighting. Island Press, Washington, DC; 2006.
- 19 Kyba C, Mohar A, Posch T. 2017. How bright is moonlight? *Astron Geophys*. 2017;58(1):31-2.
- 20 Illuminating Engineering Society. IES TM-18-18, Light and Human Health: An Overview of the Impact of Optical Radiation on Visual, Circadian, Neuroendocrine, and Neurobehavioral Responses. New York: IES; 2018.

- 21 Illuminating Engineering Society. ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: [www.ies.org/standards/definitions/](http://www.ies.org/standards/definitions/). (Accessed 2019 Jun 26)).
- 22 Longcore T, Rich C. Ecological light pollution. *Frontiers Ecol Environ*. 2004;2:191-8.
- 23 Hölker F, Wolter C, Perkin EK, Tockner K. Light pollution as a biodiversity threat. *Trends Ecol Evolution*. 2010;25:681–682.
- 24 Gaston, K. J., T. W. Davies, J. Bennie, and J. Hopkins. 2012. Reducing the ecological consequences of night-time light pollution: options and developments. *J Appl Ecol*. 2012;49:1256-66.
- 25 Longcore T, Rich C. Artificial Night Lighting and Protected Lands: Ecological effects and management approaches (Rev Aug 2017). Fort Collins, Colo: National Park Service; 2017. (NPS/NRSS/NSNS/NRR—2017/1493).
- 26 Salmon M. Artificial night lighting and turtles. *Biologist*. 2003;50:163-8.
- 27 La Sorte FA, Fink D, Buler JJ, Farnsworth A, Cabrera-Cruz SA. 2017. Seasonal associations with urban light pollution for nocturnally migrating bird populations. *Global Change Biol*. 2017;23:4609-19.
- 28 Klem D. Preventing bird-window collisions. *Wilson J Ornithology*. 2009;121(2):314-21.
- 29 Longcore T, Rich C, Gauthreaux Jr, SA. Height, guy wires, and steady-burning lights increase hazard of communication towers to nocturnal migrants: A review and meta-analysis. *Auk*. 2008;125(2):485-92.
- 30 Briggs WR. Physiology of plant responses to artificial lighting. In: Rich C, Longcore T, ed., *Ecological Consequences of Artificial Night Lighting*. Washington, DC: Island Press; 2006:389–411.
- 31 Bennie J, Davies TW, Cruse D, Gaston KJ. 2016. Ecological effects of artificial light at night on wild plants. *J Ecol*. 2016;104:611-20.
- 32 Palmer M, Gibbons R, Bhagavathula R, Holshouser D. 2018. Roadway Lighting's Impact on Altering Soybean Growth – Vol 2: LED versus HPS Color Spectral Impact, A report of the findings of ICT PROJECT R27-172. Rantoul, IL: Illinois Center for Transportation. (FHWA-ICT-18-009).
- 33 Viera-Pérez M, Hernández-Calvento L, Hesp P, Santana-del Pino A. Effects of artificial light on flowering of foredune vegetation. *Ecology*. 2019;100:e02678.
- 34 Chaney WR. (2002). *Does Night Lighting Harm Trees?* West Lafayette, IN: Purdue University; Forestry and Natural Resources. (FNR-FAQ-17).
- 35 Massetti L. Assessing the impact of street lighting on *Platanus x acerifolia* phenology. *Urban Forestry Urban Greening*. 2018;34:71-7.
- 36 Knop E, Zoller L, Ryser R, Gerpe C, Hörler M, Fontaine C. Artificial light at night as a new threat to pollination. *Nature*. 2017;548:206-9.
- 37 Longcore T, Rodriguez A, Witherington B, Penniman JF, Herf L, Herf M. Rapid assessment of lamp spectrum to quantify ecological effects of light at night. *J Exp Zool Part A: Ecol Integrat Physiol*. 2018; 329:511-21.
- 38 Wikimedia Commons. Online: [https://commons.wikimedia.org/wiki/File:Effet\\_de\\_la\\_pollution\\_lumineuse\\_sur\\_l%27arbre\\_urbain\\_Effet\\_of\\_light\\_pollution\\_on\\_urban\\_trees\\_and\\_dead\\_leaves\\_03.jpg](https://commons.wikimedia.org/wiki/File:Effet_de_la_pollution_lumineuse_sur_l%27arbre_urbain_Effet_of_light_pollution_on_urban_trees_and_dead_leaves_03.jpg). (Accessed 2019 Oct 28).
- 39 Falchi F, Cinzano P, Duriscoe D, Kyba CCM, Elvidge CD, Baugh K, Portnov BA, Rybnikova NA, Furgoni R. The new world atlas of artificial night sky brightness. *Science Advances*. 2016;2(6):e1600377. DOI: 10.1126/sciadv.1600377.

- 40 Hänel A, Posch T, Ribas SJ, Aubé M, Duriscoe D, Jechow A, Kollath Z, Lolkema DE, Moore C, Schmidt N, Spoelstra H, Wuchterl G, Kyba CCM. Measuring night sky brightness: Methods and challenges. *J Quantitative Spectroscopy Radiative Transfer*. 2017 Sep. DOI: 10.1016/j.jqsrt.2017.09.008.
- 41 Kocifaj M, Kundracik F. Modeling the night sky brightness distribution via new SkyGlow Simulator. *IEEE Lighting Conf Visegrad*; 2016. DOI:10.1109/LUMENV.2016.7745553.
- 42 Kinzey B, Perrin T, Miller N, Kocifaj M, Aubé M, Lamphar HA. An Investigation of LED Street Lighting's Impact on Sky Glow. Washington, DC: U.S. Department of Energy; 2017. DOE sky glow tool may be obtained from: <https://www.energy.gov/eere/ssl/potential-impacts-led-street-lighting-sky-glow>. (Accessed Jun 26).
- 43 Aubé M, Simoneau A. New features to the night sky radiance model illumina: Hyperspectral support, improved obstacles and cloud reflection. *J Quantitative Spectroscopy Radiative Transfer* 2018;211:25-34. DOI: 10.1016/j.jqsrt.2018.02.033.
- 44 Wikimedia Commons. Online: [https://commons.wikimedia.org/wiki/File:Griffith\\_Observatory\\_2012\\_Light\\_pollution.jpg](https://commons.wikimedia.org/wiki/File:Griffith_Observatory_2012_Light_pollution.jpg). (Accessed 2020 Jan 6).
- 45 Illuminating Engineering Society. ANSI/IES RP-39-19, Recommended Practice: Off-Roadway Sign Luminance. New York: IES; 2019.
- 46 Illuminating Engineering Society. ANSI/IES TM-15-20, Technical Memorandum: Luminaire Classification System for Outdoor Luminaires. New York: IES; 2020.
- 47 Illuminating Engineering Society. IES G-1-16, Security Lighting for People, Property, and Critical Infrastructure. New York: IES; 2016.
- 48 Sivak M, Schoettle B, Minoda T, Flanagan MJ. Blue Content of LED Headlamps and Discomfort Glare. Ann Arbor: Univ Michigan; 2002. (Report No. UMTRI 2002-2).
- 49 Bullough JD, Fu Z, Van Derlofske J. Discomfort and disability glare from halogen and HID headlamp systems. In: *Advanced Lighting Technology for Vehicle*, SP-1668. Warrendale, Penn.: Society of Automotive Engineers; 2002:1-5. (SAE Paper 2002-01-0010).
- 50 Illuminating Engineering Society. ANSI/IES LM-63-19, Standard File Format for Electronic Transfer of Photometric Data. New York: IES; 2019.
- 51 Illuminating Engineering Society. ANSI/IES TM-33-18, Standard Format for the Electronic Transfer of Luminaire Optical Data. New York: IES; 2018.

## **Process for Change to an ANSI/IES Standard under Continuous Maintenance**

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

### **Submittal Format**

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

### **Submission to the Committee Chair**

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

### **Review and Clarification**

Responders shall review proposals and should contact the proposer if necessary for clarification.

### **Response Recommendation**

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

### **Editing**

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

## Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17<sup>th</sup> Floor, New York, NY 10005-4001. Email: [standards@ies.org](mailto:standards@ies.org). Fax: 212-248-5017.

1. Submitter: \_\_\_\_\_  
Affiliation: \_\_\_\_\_  
Address: \_\_\_\_\_  
City: \_\_\_\_\_ State: \_\_\_\_\_ Zip: \_\_\_\_\_ Country: \_\_\_\_\_  
Telephone: \_\_\_\_\_  
Fax: \_\_\_\_\_  
E-mail: \_\_\_\_\_

I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: \_\_\_\_\_ Date: \_\_\_\_\_

2. Title of publications and year published \_\_\_\_\_

3. Clause (section), sub-clause or paragraph number; and page number: \_\_\_\_\_

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

---

---

---

6. Reason and substantiation:

---

---

---

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: \_\_\_\_\_
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:





### **Lighting Science Standards**

Fundamentals, Metrics and Calculations

### **Lighting Practice Standards**

Design, Engineering, and Specifications

### **Lighting Applications Standards**

Design Criteria and Illumination Recommendations

### **Lighting Measurements and Testing Procedure Standards**

Industry Standardization

### **Roadway and Parking Facility Lighting Standards**

Criteria and Illumination Recommendations

**Order# ANSI/IES LP-11-20**

**ISBN# 978-0-87995-354-6**

**[www.ies.org](http://www.ies.org)**